

CAPE ROSS, Antarctica

WMO: 896660

Lat: 76.715S

Lon: 162.970E

Elev: 495

StdP: 14.43

Time Zone: 12.00 (NZL)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-30.3	-28.0	N/A	N/A	N/A	N/A	N/A	N/A	52.6	1.4	46.4	-3.8	12.8	170	1.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	35.8	N/A	33.9	N/A	32.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
45.2	38.4	32.7	-35.0	40.0	3.7	2.9	-37.6	42.0	-39.8	43.7	-41.9	45.3	-44.6	47.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	27.5	17.9	5.2	-3.1	-4.8	-6.5	-12.1	-8.6	-7.3	1.9	15.8	26.1		
	DBStd	15.68	3.72	5.44	7.07	8.61	10.18	9.58	10.05	10.32	9.77	6.70	6.99	4.11		
	HDD50	16689	698	899	1388	1594	1698	1695	1924	1818	1718	1492	1026	739		
	HDD65	22164	1163	1319	1853	2044	2163	2145	2389	2283	2168	1957	1476	1204		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.1	10.3	15.2	15.9	14.7	15.1	16.5	14.4	16.3	16.4	13.9	11.3	9.8		
	PrecAvg	8.3	0.9	0.9	0.7	0.5	0.6	0.7	0.5	0.7	0.7	0.7	0.7	0.7		
Precipitation	PrecMax	18.5	1.9	2.4	2.7	1.4	1.9	2.4	1.5	2.9	2.3	2.4	2.0	2.4		
	PrecMin	1.9	0.1	0.0	0.0	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	4.3	0.6	0.7	0.6	0.3	0.5	0.6	0.4	0.7	0.5	0.6	0.4	0.6		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	31.5	21.9	21.9	19.7	17.8	16.7	17.3	16.7	23.5	31.3	39.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	36.0	28.7	19.3	15.4	15.5	14.0	11.7	12.5	12.9	16.6	28.6	36.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	34.2	26.9	17.1	12.3	12.5	10.6	6.9	8.8	9.9	13.8	26.9	34.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	32.7	25.0	14.6	8.8	9.5	7.0	2.8	5.8	6.4	11.5	25.0	32.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	MDBR	5.8	5.9	6.2	8.2	8.7	9.2	9.9	10.9	10.2	8.7	7.8	6.4		
		MCDBR	7.3	7.4	8.4	10.6	10.4	11.9	13.2	13.0	12.8	10.9	9.4	8.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Clear-Sky Solar Irradiance	taub	0.236	0.224	0.179	0.111	N/A	N/A	N/A	N/A	0.109	0.167	0.197	0.218	0.234		
	taud	2.305	2.320	2.229	1.906	N/A	N/A	N/A	N/A	1.888	2.172	2.278	2.295	2.302		
	Ebn at Noon	309	287	251	75	0	0	0	0	69	252	299	316	317		
	Edh at Noon	30	23	15	2	0	0	0	0	2	15	24	30	32		
All-Sky Solar Radiation	RadAvg	2659	1578	583	62	0	0	0	0	12	329	1234	2444	3062		
	RadStd	64	51	25	3	0	0	0	0	1	16	37	64	79		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page